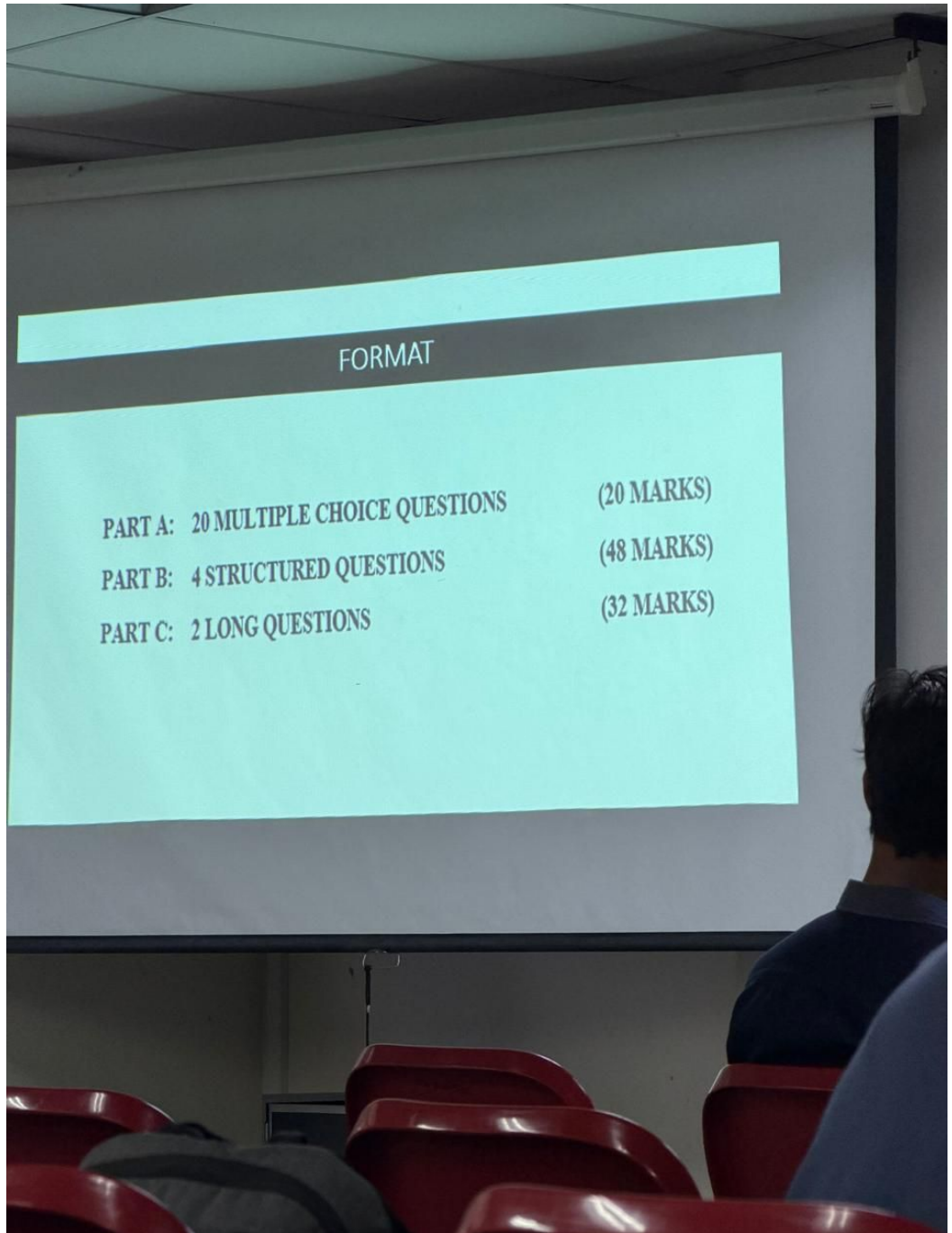
SCSM2113 Final exam 25/26-1

- 5th February, Thursday
  - 9am – 12 noon
  - @DK8, N24
- Questions format:
  - Multiple choice questions (all topics)
  - Structured questions (topic Design till Evaluation)
  - Problem analysis questions – long question (all topics but leaning more towards Design till end)

  Total questions:
  - **15 MCQ, 3 structured, 2 long questions**
  -


FORMAT
PART A: 20 MULTIPLE CHOICE QUESTIONS (20 MARKS)
PART B: 4 STRUCTURED QUESTIONS (48 MARKS)
PART C: 2 LONG QUESTIONS (32 MARKS)

HCI TOPICS COVER

- All topics included, but may emphasize on:

Topic	Title	Contents
3	Interaction Conceptualising Design and Trends in Interfaces	• Conceptualising Design and Problem Space • interface metaphors & interaction type
4	The Process of Interaction Design	• User-Centered Design approach • Double Diamond Model • User involvement* • Agile development Basic Concepts (Sprints)*
5	Establishing Requirements and Data gathering	• Definition Scenario, persona* • Data Gathering Method. Eg contextual inquiry etc • Hierarchical Task Analysis (HTA), use case • User Requirements Types and Why importance
6	Design, Prototyping and Construction User Journey Map	• Purpose Prototypes,* Gestalt Principles applied in the UI design* • Definition Conceptual design & Concrete design * • Storyboard, user journey map*
7 & 8	Introducing Evaluation & Other evaluation types	• Evaluation Methods – Control Setting (Usability Testing & Living Lab), Natural Setting (In the wild), A/B Testing • Usability heuristic evaluation and the severity usability problem

PART B : 4 SHORT STRUCTURE QUESTIONS

Question 2 **(13 Marks)**

a) List and briefly describe the **TWO (2)** interaction types. Provide examples for each interaction type. (7 Marks)

b) Define the term "Interface metaphor" and explain its purpose in interface design. Provide **TWO (2)** examples of how it is used in modern systems. (6 Marks)

PART B : 4 SHORT STRUCTURE QUESTIONS

Question 4: **[15 marks]**

Figure 2 (refer page 5) shows the Natural History Museum website. Based on Figure 2, answer the following questions.

a. List down FIVE (5) heuristic evaluation that you have learned in class. [5 marks]

b. Identify and explain THREE (3) examples of Nielsen's usability heuristic found in Figure 2. [6 marks]

c. Explain TWO (2) examples how the Gestalt Principles are used to group things in Figure 2. [4 marks]

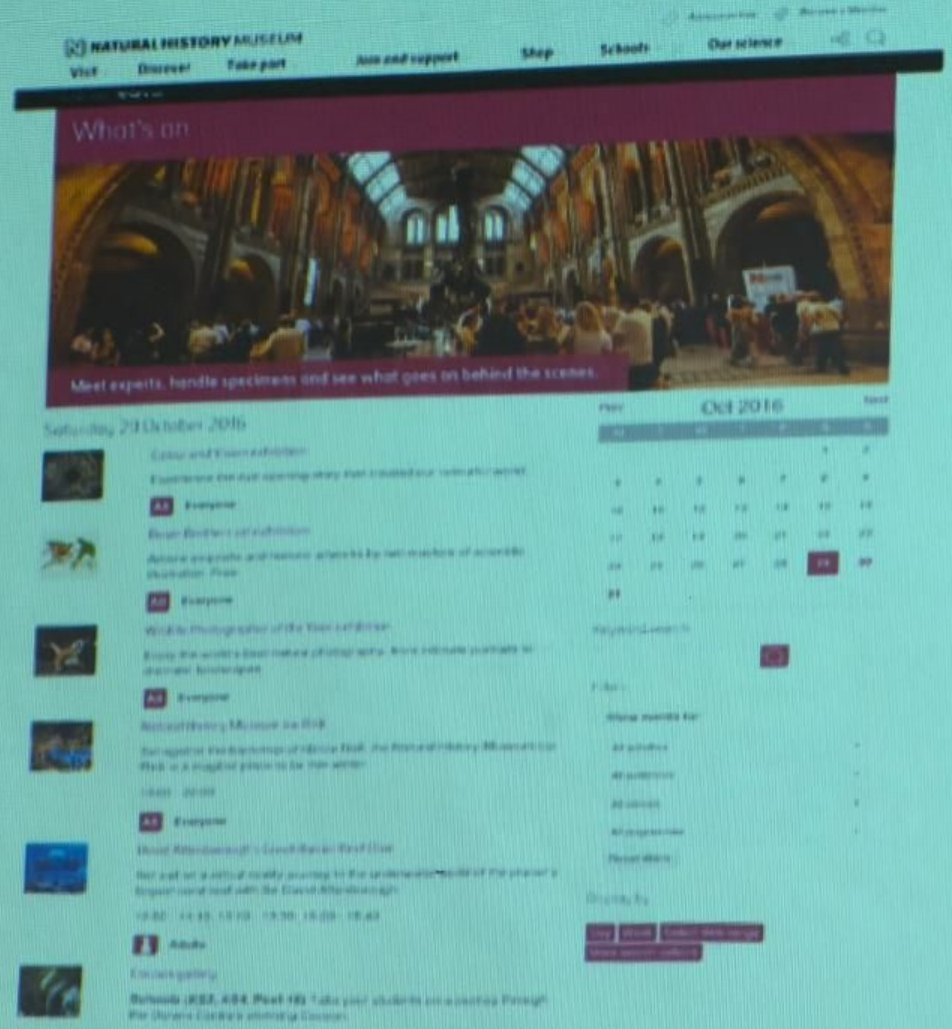


Figure 2. Natural History Museum website

PART C : 2 LONG STRUCTURE QUESTIONS

[16 Marks]

Question 2:

Read the following description.

Splendid Hotels' have been in the accommodation business for approximately five years. In 2002, they launched their first website which comprised a very basic interface for advertising their rooms and services. Last year, a new online room booking facility was built into their website in hope to alleviate bookings made via their phone service. Since launching this new service, annual reports revealed that customers continued to make bookings via phone and have discovered that customers preferred to book via phone because the website was too confusing and difficult for them to use. Consequently, Splendid Hotels have sought to hire your team as a usability consultant to re-design their Online Bookings screen to make it more 'user-friendly' and evaluate the prototype design.

a. What is the goal of an evaluation? [2 marks]
b. Describe THREE (3) types of evaluation setting environment by giving each an example. [6 marks]
c. Create a usability test plan (step-by-step) that describes how you would conduct a *usability testing* of the prototype design for the new Online Room Booking? [6 marks]
d. Describe ONE (1) example of parameter in measuring efficiency for the task of Online Room Booking facility. [2 marks]

[15 marks]

Question 2

You are hired as a consultant to design an interactive product in a self-service cafeteria that allows users to pay for their food using a debit / credit system.

a. Based on the scenario above, suggest some key requirements needed to design the interactive self-service product at the cafeteria in each the following category. [10 marks]

   i. Functional requirements
   ii. Data requirements
   iii. User Characteristics
   iv. Usability goal
   v. User experience goals

b. Suggest and elaborate the **TWO (2)** suitable usability heuristic design guideline that you think appropriate to implement in the interactive product at the cafeteria in order to achieve the goals. [5 marks]

Case Study 1

- **Identifying Usability Issues.** How you would identify the usability problems in this Case Study
  1. What Method chosen : Usability testing / HE / Contextual Inquiry
  2. Reasoning for method Chosen
  3. Types of Issues identified

- Propose the system **improvement plan.**
  1. Action taken to fix/reduce the identified issues
  2. Justify why the action propose is appropriate
  3. How to check whether the improvement is ok or not

PART B : 4 SHORT STRUCTURE QUESTIONS

Format Question:

a) Define the term "……",

e.g Define the term usability goals

b) List and briefly describe **FOUR (4)** "……" .

*e.g List and briefly describe **FOUR (4)** usability goals*

c) Provide **TWO (2)** examples of "……" .

*e.g Provide **TWO (2)** examples of User experience goals*